```
****************************************************
*                                                  *
*           STAAD.Pro                              *
*           Version  2007    Build 04              *
*           Proprietary Program of                 *
*           Research Engineers,  Intl.             *
*           Date=    DEC 18, 2012                  *
*           Time=    21:28:12                      *
*                                                  *
*      USER ID: Parsons Brinckerhoff               *
****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: 121217 - TIE BEAM TB1.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 22-NOV-11
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 3 0 0 0; 4 7.6 0 0; 5 -6.4 0 0; 6 13.7 0 0
     9. MEMBER INCIDENCES
    10. 3 3 4; 4 3 5; 5 4 6
    11. DEFINE MATERIAL START
    12. ISOTROPIC CONCRETE
    13. E 2.17185E+007
    14. POISSON 0.17
    15. DENSITY 23.5616
    16. ALPHA 1E-005
    17. DAMP 0.05
    18. END DEFINE MATERIAL
    19. MEMBER PROPERTY AMERICAN
    20. 3 TO 5 PRIS YD 0.8 ZD 0.7
    21. CONSTANTS
    22. MATERIAL CONCRETE ALL
    23. SUPPORTS
    24. 3 TO 5 PINNED
    25. 6 FIXED BUT MY MZ KMX 0.001
    26. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD
    27. SELFWEIGHT Y -1.4
    28. MEMBER LOAD
    29. 3 TO 5 CON GY -1062.1
    30. 3 TO 5 CON GZ 19
    31. 3 TO 5 CON GX 200
    32. 3 TO 5 CMOM GY 0.05
    33. 3 CMOM GX 58
    34. 3 TO 5 CMOM GZ 500
    35. 3 TO 5 UNI GZ 21
    36. ***********************
    37. *VM1 VM2 PLATFORM
    38. 3 TO 5 CON GY -155
    39. 3 TO 5 CON GZ 25
    40. 3 TO 5 CON GX 36
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    41. 3 TO 5 CMOM GY 0.1
    42. 3 CMOM GX 3
    43. 3 TO 5 CMOM GZ 46
    44. ***********************
    45. LOAD COMB 2 1.0DL
    46. 1 1.0
    47. LOAD COMB 3 1.4DL
    48. 1 1.4
    49. PERFORM ANALYSIS
```

P R O B L E M   S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =   4/   3/   4

SOLVER USED IS THE IN-CORE ADVANCED SOLVER

TOTAL PRIMARY LOAD CASES =   1, TOTAL DEGREES OF FREEDOM =   12

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    50. LOAD LIST 2 3
    51. PRINT JOINT DISPLACEMENTS ALL
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JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
3	2	0.000E+00	0.000E+00	0.000E+00	1.065E+03	3.740E-05	3.246E-04
	3	0.000E+00	0.000E+00	0.000E+00	1.491E+03	5.235E-05	4.544E-04
4	2	0.000E+00	0.000E+00	0.000E+00	1.065E+03	-1.717E-05	-7.327E-04
	3	0.000E+00	0.000E+00	0.000E+00	1.491E+03	-2.404E-05	-1.026E-03
5	2	0.000E+00	0.000E+00	0.000E+00	1.065E+03	-3.711E-04	-3.143E-03
	3	0.000E+00	0.000E+00	0.000E+00	1.491E+03	-5.196E-04	-4.401E-03
6	2	0.000E+00	0.000E+00	0.000E+00	1.065E+03	3.194E-04	2.419E-03
	3	0.000E+00	0.000E+00	0.000E+00	1.491E+03	4.471E-04	3.387E-03

************** END OF LATEST ANALYSIS RESULT **************

52. PRINT MEMBER FORCES ALL

MEMBER END FORCES STRUCTURE TYPE = SPACE

ALL UNITS ARE -- KN METE (LOCAL)

MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
3	2	3	-118.00	759.82	-102.85	0.00	150.36	1370.48
		4	-118.00	597.67	-100.75	-61.00	-142.53	-1300.31
	3	3	-165.20	1063.75	-143.99	0.00	210.50	1918.68
		4	-165.20	836.74	-141.05	-85.40	-199.55	-1820.44
4	2	3	118.00	796.49	112.67	0.00	-150.36	1370.48
		5	118.00	538.84	65.73	0.00	0.00	0.00
	3	3	165.20	1115.08	157.74	0.00	-210.50	1918.68
		5	165.20	754.37	92.02	0.00	0.00	0.00
5	2	4	-118.00	967.57	-109.44	61.00	142.53	1300.31
		6	-118.00	362.22	-62.66	-61.00	0.00	0.00
	3	4	-165.20	1354.59	-153.22	85.40	199.55	1820.44
		6	-165.20	507.10	-87.72	-85.40	0.00	0.00

************** END OF LATEST ANALYSIS RESULT **************

53. PRINT SUPPORT REACTION ALL

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
3	2	-236.00	1556.31	-215.52	0.00	0.00	0.00
	3	-330.40	2178.83	-301.73	0.00	0.00	0.00
4	2	-236.00	1565.23	-210.19	0.00	0.00	0.00
	3	-330.40	2191.33	-294.27	0.00	0.00	0.00
5	2	-118.00	538.84	-65.73	0.00	0.00	0.00
	3	-165.20	754.37	-92.02	0.00	0.00	0.00
6	2	-118.00	362.22	-62.66	-61.00	0.00	0.00
	3	-165.20	507.10	-87.72	-85.40	0.00	0.00

************** END OF LATEST ANALYSIS RESULT **************

```
  54. LOAD LIST ALL
  55. START CONCRETE DESIGN
  56. CODE CP65
PROGRAM CODE REVISION V1.0_CP65/1
  57. DESIGN BEAM 3 TO 5
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====================================================================

       B E A M  N O.    3  D E S I G N  R E S U L T S -  FLEXURE

   LEN -  7600. mm   FY - 460.  FC - 30.  SIZE -  700. X  800. mm

LEVEL    HEIGHT     BAR INFO       FROM           TO        ANCHOR
           mm                       mm            mm       STA  END
-------------------------------------------------------------------

  1       50.      5- 50 MM       742.        6224.      NO   NO
         COMP.    15- 8  MM   (REQD. STEEL=    728. SQ. MM)
  2      743.      5- 50 MM         0.        2424.      YES  NO
         COMP.    15- 8  MM   (REQD. STEEL=    728. SQ. MM)
  3      748.      6- 40 MM      4542.        7600.      NO   YES

     B E A M   N O.    3 D E S I G N  R E S U L T S -  SHEAR

           PROVIDE SHEAR AND TORSOINAL LINKS AS FOLLOWS
|------------------------------------------------------------------------|
|  FROM - TO  | SHEAR TORSN | LOAD | LINK |      NO.    | SPACING mm C/C |
|      mm     |  kN    kNm  | S  T | SIZE |  S  T  S+T  |   S    T   S+T |
|-------------|-------------|------|------|-------------|----------------|
| END 1  3799 | *****   85  |  3  3| 8 mm | 69  20  92  |   54  183   41 |
|  3799 END 2 | 836.7   85  |  3  3| 8 mm | 48  20  70  |   78  183   54 |
|------------------------------------------------------------------------|
|EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 219 mm2 EVENLY DISTRIBUTED|
|      * TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - CP65 *       |
|------------------------------------------------------------------------|

====================================================================

       B E A M  N O.    4  D E S I G N  R E S U L T S -  FLEXURE

   LEN -  6400. mm   FY - 460.  FC - 30.  SIZE -  700. X  800. mm

LEVEL    HEIGHT     BAR INFO       FROM           TO        ANCHOR
           mm                       mm            mm       STA  END
-------------------------------------------------------------------

  1       50.      5- 50 MM       976.        6400.      NO   YES
         COMP.    15- 8  MM   (REQD. STEEL=    728. SQ. MM)
  2      743.      5- 50 MM         0.        2758.      YES  NO
         COMP.    15- 8  MM   (REQD. STEEL=    728. SQ. MM)
```